UNISYNC 2.0

Complete UI/UX and Functional Upgrade Plan

Skill Development Project – Repeat / Upgrade

Project scope
Mobile Application + Web Application

Backend
Firebase

Redesign approach
Hybrid Redesign – preserve useful existing elements and improve the user experience and visual identity.

Prepared as a practical roadmap for improving the existing UniSync project and demonstrating clear design, usability and functional progress.

Project Overview

The purpose of UniSync 2.0 is to upgrade the UI/UX work completed for the previous Skill Development Project. The new version should show that the existing product was reviewed, usability problems were identified, and meaningful improvements were made. This should be more than a visual refresh: it should demonstrate a stronger user experience and a working application supported by Firebase.

Area	Recommended direction
Platforms	Mobile application and responsive web application
Redesign approach	Hybrid redesign: keep useful existing elements and improve weak areas
Backend	Firebase
User roles	Student, Lecturer and Admin
Main outcome	A professional, usable academic management platform with documented before-and-after improvements

1. UI Design Improvements

Current area	Changes to make
Colour palette	Use a consistent colour system. Retain the existing purple accent if it is part of the UniSync identity.
Typography	Use a consistent typeface such as Inter, with clear font sizes and weights.
Buttons	Standardise button sizes, colours, labels and interaction states.
Cards and spacing	Improve card layouts, alignment, spacing and visual hierarchy.
Forms	Improve input fields, labels, validation and error messages.
Icons	Use one consistent icon style.
Navigation	Make navigation simpler and consistent across mobile and web.
Dark mode	Improve contrast and readability if dark mode is retained.
Responsive design	Optimise layouts for mobile, tablet and desktop.
System feedback	Add useful empty states, loading indicators and success/error feedback.

Suggested visual direction

Navy #111827 · Purple #6D4AFF · Background #F5F6FA · White #FFFFFF · Success #16A36A · Warning #F59E0B · Danger #E5484D

These colours are a proposal. Compare them with the original UniSync design before finalising the design system.

2. UX Improvements

Area	Recommended improvement
User journey	Reduce unnecessary steps in common tasks.

Area	Recommended improvement
Navigation	Make important features easy to find.
Dashboard	Show the most important academic information first.
Task management	Make assignment tracking and submission easier.
Feedback	Provide clear success, warning and error messages.
Accessibility	Improve contrast, text readability and touch-target sizes.
Consistency	Use familiar interaction patterns across screens.
Personalisation	Display information relevant to the signed-in user's role.

Example UX goal: a student should be able to see upcoming assignment deadlines immediately after logging in, without opening multiple pages.

3. Functional Feature Improvements

A. Student features

Feature	Recommended functionality
Smart Student Dashboard	Upcoming deadlines, today's schedule, pending assignments, GPA overview, course progress and recent notifications.
Assignment Management	View by course; add to planner; set priority; track Pending / In Progress / Completed; upload files; show deadlines.
Academic Calendar	Monthly and weekly views; assignment deadlines; lectures; examinations; academic events.
Advanced GPA Calculator	Add subjects and credits; select grades; calculate semester GPA and cumulative GPA; view performance history.
Study Planner	Create study sessions; set daily goals; track completed sessions; view weekly progress; set reminders.
Course Management	View enrolled courses, course details, lecturer information, related assignments and course progress.

B. Lecturer features

- Create and publish assignments, including deadlines.
- Upload learning materials and manage assigned courses.
- View student submissions and submission status.
- Review submissions and provide feedback or marks if included in the project scope.
- Publish course announcements and send notifications to students.

C. Admin features

- Manage student and lecturer accounts.
- Add, edit and remove courses.
- Assign lecturers to courses.
- Manage academic information and announcements.
- View system activity and high-level system statistics.

Role permissions: Student, Lecturer and Admin must have separate permissions. A student must not be able to access lecturer or admin functions. Enforce these permissions in Firebase Security Rules and in the application UI.

4. Mobile Application Improvements

- Use bottom navigation for the main sections.
- Keep the dashboard simple, readable and focused on quick actions.
- Use floating action buttons only where they make common actions easier.
- Make forms and controls comfortable to use on small screens.
- Add push notifications for relevant reminders.
- Support different mobile screen sizes and orientations where appropriate.

Suggested mobile bottom navigation: Home | Assignments | Calendar | GPA | Profile

5. Web Application Improvements

- Use a left sidebar for primary navigation.
- Use a top bar for search, notifications and account access.
- Show summary cards and relevant academic information on the dashboard.
- Use data tables with sorting and filtering where useful.
- Provide detailed assignment, calendar and GPA views.
- Use separate dashboards and navigation for each user role.

6. Firebase Backend Improvements

Firebase service	Purpose
Firebase Authentication	Login, registration and authentication.
Cloud Firestore	Store users, courses, assignments, submissions, study sessions and notifications.
Cloud Storage	Store assignment documents and profile images.
Firebase Cloud Messaging	Deliver push notifications.
Firestore Security Rules	Restrict access according to authenticated users and their roles.

Suggested Cloud Firestore structure

users/{userId}: name, email, role, studentId
courses/{courseId}: courseName, courseCode, lecturerId
assignments/{assignmentId}: title, courseId, deadline, status
submissions/{submissionId}: studentId, assignmentId, fileUrl, submittedAt
studySessions/{sessionId}: studentId, subject, date, duration
notifications/{notificationId}: userId, title, message, isRead

This is a starting structure, not a final database design. Define relationships, ownership, indexes, validation and security rules before implementation. Do not rely only on hiding buttons in the UI to protect data.

7. Recommended Screen List

Screen	Mobile	Web
Splash / Login / Register	Yes	Login / Register
Student Dashboard	Yes	Yes
Assignment List / Details	Yes	Yes
Add / Edit Assignment	Yes	Yes
Academic Calendar	Yes	Yes
GPA Calculator	Yes	Yes
Study Planner	Yes	Yes
Course List / Details	Yes	Yes
Notifications	Yes	Yes
Profile / Settings	Yes	Yes
Lecturer Dashboard	Yes	Yes
Lecturer Assignment Management	Yes	Yes
Student Submissions	Yes	Yes
Admin Dashboard	Yes	Yes
User Management	Yes	Yes

Screen	Mobile	Web
Course Management	Yes	Yes

Prioritise the screens for features that will actually be implemented. Not every screen needs the same level of detail in the prototype.

8. What Will Make the Repeated Project Stronger?

For the evaluation, demonstrate the reasoning behind the changes. Do not present only the final screens. Show what was wrong, what you changed and how the new design improves the experience.

Deliverable	What to demonstrate
Existing design	Original UniSync mobile and web screens.
UX audit	Usability and visual problems found in the original design.
User flows	Improved journeys for common student, lecturer and admin tasks.
Wireframes	New screen structures before high-fidelity design.
Design system	Colours, typography, spacing, components and states.
High-fidelity UI	Final mobile and web designs.
Prototype	Clickable flows and interactions.
Firebase integration	Working authentication, database and relevant features.
Testing	Feedback from users and issues found during testing.
Before vs After	Clear comparison and explanation of improvements.

9. Recommended Implementation Order

Phase 1 – Existing project analysis

- Review the previous mobile UI and web UI.
- Identify usability and visual problems.
- Prepare a before-and-after comparison plan.

Phase 2 – UX and UI redesign

- Create user flows and improve information architecture.
- Create the design system and reusable components.
- Redesign the student dashboard, assignment management, calendar and GPA screens.

Phase 3 – Role-based features

- Design lecturer and admin dashboards.
- Create role-specific navigation and permissions.
- Design course and user management.

Phase 4 – Firebase integration

- Implement authentication and user roles.
- Create Firestore collections and data access patterns.
- Implement assignment and submission management.
- Add file storage, notifications and security rules.

Phase 5 – Testing and demonstration

- Test mobile and web responsiveness.
- Test role permissions and core workflows.
- Conduct usability testing and document findings.
- Prepare the project report and presentation.

10. Five Highest-Priority Upgrades

Priority	Upgrade	Why it matters
1	Improved Student Dashboard	Creates a clear starting point for academic activities.
2	Advanced Assignment Management	Addresses deadlines, priorities, progress and submissions.
3	Lecturer and Admin Dashboards	Demonstrates role-based functionality and broader system value.
4	Consistent Mobile + Web Design	Shows a coherent experience across platforms.
5	Firebase Integration	Connects the design to a functioning application.

Final recommendation: Do not add features simply to make the project bigger. Choose features that solve real problems for students, lecturers or administrators. Audit the original UniSync designs first, then decide what to keep, improve or add.